

Analog Circuits Tutorial

Transistor Bias, Amplifiers, and Frequency Response

Prepared by: Ameenudeen PE, Asst. Professor, ECE

Structure: 30 questions comprising 20 theory/viva questions and 10 numerical problems.

Coverage: BJT basics, bias and operating regions, load lines, large- and small-signal models, CE/CB/CC amplifiers, amplifier design, cascade and cascode configurations, RC-coupled frequency response, and Class A/B power amplifiers.

Progression: Questions move from basic concepts to analysis, design, and laboratory interpretation.

General instructions

- Draw circuit diagrams or equivalent circuits wherever useful.
- State assumptions before using an approximate formula.
- For bias calculations, verify that the calculated Q-point lies in the intended operating region.
- Distinguish DC values, peak AC amplitudes, and RMS values.
- Use $V_{BE} \approx 0.7\text{ V}$, $V_{CE(\text{sat})} \approx 0.2\text{ V}$, and $V_T = 25\text{ mV}$ unless otherwise specified.
- Neglect the Early effect unless r_o or V_A is given.
- Treat coupling and bypass capacitors as shorts only when the stated frequency range permits that approximation.

Part A: Theory and viva questions (20 questions)

Q1. BJT basics — Basic

- (a) Identify the emitter, base, and collector of an NPN transistor. Explain why their physical construction is not identical.
- (b) Explain the roles of emitter injection and the thin base region in transistor action.
- (c) State the relationships among I_E , I_C , and I_B . Define α and β , and derive their relationship.

Q2. Operating regions — Basic

Explain cutoff, forward-active operation, and saturation using the bias conditions of the base-emitter and base-collector junctions. Which region is normally used for linear amplification, and why does $I_C \approx \beta I_B$ cease to be reliable in saturation?

Q3. Understanding transistor voltages — Basic

Explain why $V_{BE} \approx 0.7\text{ V}$ and $V_{CE(\text{sat})} \approx 0.2\text{ V}$ are approximations rather than universal constants. Use $V_{CE} = V_{BE} - V_{BC}$ to explain the small collector-emitter voltage in saturation.

Q4. Bias, Q-point, and swing — Basic

Define bias, Q-point, peak signal amplitude, and peak-to-peak swing. Explain why a transistor can carry a nonzero collector current when the input AC signal is zero. Discuss what happens when the Q-point is placed too close to cutoff or saturation.

Q5. DC and AC load lines — Intermediate

Derive the DC load line for a CE amplifier with R_C and R_E . Then explain why an AC-coupled load and an emitter-bypass capacitor can change the AC load line without changing the DC load line. Why does the AC load line pass through the Q-point?

Q6. Fixed bias versus voltage-divider bias — Intermediate

Draw both bias circuits and derive the expressions for their base currents. Explain why fixed bias is sensitive to transistor β . Use the Thevenin equivalent of a voltage divider to explain base-current loading and the benefit of an emitter resistor.

Q7. Bias stability and temperature — Intermediate

Explain how an emitter resistor opposes an increase in collector current. Discuss how temperature can affect V_{BE} , leakage current, and the Q-point. What does “thermal runaway” mean, and how can circuit design reduce its likelihood?

Q8. Large-signal and small-signal models — Intermediate

Starting with $I_C \approx I_S e^{V_{BE}/V_T}$, derive $i_c \approx g_m v_{be}$. Explain why the Q-point must be known before calculating g_m , and why this linearized model cannot predict cutoff or saturation accurately.

Q9. Physical intuition for the hybrid- π model — Intermediate

Explain why the small-signal model contains:

- (a) r_π between base and emitter;
- (b) a controlled current source $g_m v_{be}$;
- (c) an optional resistance r_o between collector and emitter.

Derive $r_\pi = \beta/g_m$. Explain why r_π is not simply V_{BEQ}/I_{BQ} .

Q10. CE gain and emitter degeneration — Intermediate

Derive the base-to-collector voltage gain of a CE stage:

- (a) with its emitter at AC ground;
- (b) with an unbypassed emitter resistor.

Explain the phase inversion and the trade-off between gain, linearity, and input resistance when R_E is unbypassed.

Q11. Input and output resistance — Intermediate

Define amplifier input and output resistance. Explain how a test source can be used to determine each. When finding output resistance, why must independent signal sources be set to

zero, while controlled sources remain in the model? Explain how source resistance changes the overall source-to-output gain.

Q12. Common-base amplifier — Intermediate

Draw the small-signal circuit and derive the voltage gain and input resistance of a CB amplifier with its base at AC ground. Explain why its voltage gain is positive and its input resistance is low. Distinguish collector-current gain from voltage gain.

Q13. Emitter follower — Intermediate

Derive the voltage gain and base input resistance of a CC amplifier. Explain why its voltage gain is slightly below unity and why it is still useful as an amplifier. Discuss how source resistance affects the output resistance seen at the emitter.

Q14. Selecting an amplifier configuration — Application

Select a suitable configuration, or combination of configurations, for each requirement and justify your choice:

- (a) substantial voltage gain from a moderate-resistance source;
- (b) buffering a high-resistance source before a low-resistance load;
- (c) accepting a signal from a low-resistance source while obtaining voltage gain.

Discuss the loading and phase behavior of your choices.

Q15. Cascade amplifiers — Advanced

Explain how two RC-coupled amplifier stages are cascaded. Derive the overall gain as the product of the *loaded* stage gains and any input attenuation. Why is multiplying the unloaded gains usually incorrect? What is the overall phase relationship for two cascaded CE stages?

Q16. Cascode amplifier — Advanced

Distinguish *cascade* from *cascode*. Explain the operation of a CE–CB cascode: which transistor accepts the input and which holds the lower transistor’s collector voltage approximately constant? Discuss the Miller effect, bandwidth, output resistance, and the voltage headroom required by the two transistors.

Q17. Low-frequency response of an RC-coupled amplifier — Intermediate

Explain the roles of the input coupling, output coupling, and emitter-bypass capacitors. Identify the resistance seen by each capacitor when estimating its corner frequency. Why does gain fall at low frequencies? When is taking the largest individual corner frequency a reasonable estimate of the lower cutoff?

Q18. High-frequency response and bandwidth — Advanced

Explain how C_π , C_μ , wiring capacitance, and load capacitance affect the upper cutoff. Derive the input Miller capacitance $C_\mu(1 - A_v)$ for an inverting stage. Define midband gain, the -3 dB cutoff frequencies, and bandwidth. Describe how you would measure them in the laboratory.

Q19. Class A power amplification — Intermediate

Derive the ideal maximum efficiency of:

- (a) a series-fed resistive-load Class A amplifier;

- (b) a transformer-coupled Class A amplifier.

Explain why the maximum current amplitude is approximately I_{CQ} , why transformer coupling permits collector voltage to rise above V_{CC} , and why Class A consumes power at idle.

Q20. Class B/AB operation and laboratory diagnosis — Application

Explain conduction angle, push-pull operation, crossover distortion, and the role of Class AB bias. Derive the ideal maximum efficiency of Class B operation.

A measured amplifier output shows either flattened peaks or a small notch near its zero crossing. Explain how you would distinguish clipping from crossover distortion and identify the likely causes.

Part B: Numerical and design problems (10 questions)

Q21. Fixed bias and maximum swing — Basic numerical

An NPN fixed-bias CE amplifier has

$$V_{CC} = 12\text{ V}, \quad R_B = 200\text{ k}\Omega, \quad R_C = 1\text{ k}\Omega, \quad \beta = 100.$$

The emitter is grounded and there is no external AC load.

- Calculate I_{BQ} , I_{CQ} , and V_{CEQ} . Verify the assumed operating region.
- Find the approximate cutoff and saturation endpoints of the DC load line.
- Determine the maximum approximately symmetric collector-voltage peak swing allowed by these limits.
- Repeat the Q-point calculation for $\beta = 150$. Explain which side of the signal swing is now more restricted.

Q22. Voltage-divider bias — Intermediate numerical

A voltage-divider-biased CE amplifier has

$$\begin{aligned} V_{CC} &= 12\text{ V}, & R_1 &= 39\text{ k}\Omega, & R_2 &= 12\text{ k}\Omega, \\ R_C &= 1\text{ k}\Omega, & R_E &= 470\text{ }\Omega, & \beta &= 100. \end{aligned}$$

Here R_1 connects supply to base and R_2 connects base to ground.

- Find V_{TH} and R_{TH} .
- Calculate I_B , I_C , I_E , V_B , V_E , V_C , and V_{CE} , retaining base-current loading.
- Compare V_B with the unloaded divider voltage.
- Recalculate I_C for $\beta = 150$, and compare its percentage change with that of a fixed-bias circuit.

Q23. Design a CE bias network — Design numerical

Design a voltage-divider-biased CE stage for

$$V_{CC} = 12\text{ V}, \quad I_{CQ} = 2\text{ mA}, \quad V_{CEQ} = 6\text{ V}, \quad V_{EQ} = 1.2\text{ V},$$

assuming $\beta = 100$.

- Calculate I_{BQ} , I_{EQ} , R_E , and R_C .

- (b) Set the DC current through R_2 equal to $10I_{BQ}$. Using the required base voltage, calculate R_2 , then R_1 . Include the base current when finding the current through R_1 .
- (c) Verify the design using the Thevenin bias equation.
- (d) Choose nearby E12 resistor values and recalculate the resulting Q-point. State the resulting design error.

Q24. CE small-signal gain and resistances — Intermediate numerical

A CE stage operates at $I_{CQ} = 2 \text{ mA}$ with

$$\beta = 100, \quad V_A = 100 \text{ V}, \quad R_C = 3.3 \text{ k}\Omega, \quad R_L = 10 \text{ k}\Omega.$$

Its bias network presents $R_B = R_1 \parallel R_2 = 20 \text{ k}\Omega$, and the source resistance is $R_S = 1 \text{ k}\Omega$. The emitter bias resistor is $R_E = 470 \Omega$.

- (a) Calculate g_m , r_π , and r_o .
- (b) With R_E fully bypassed at midband, calculate the base-to-output gain, including r_o .
- (c) Find the total input resistance, source-to-output gain, and unloaded output resistance for the bypassed case.
- (d) With R_E unbypassed, neglect r_o and calculate the new base-to-output gain and input resistance. Explain the changes.

Q25. Emitter follower and load driving — Intermediate numerical

An emitter follower operates at $I_{CQ} = 2 \text{ mA}$ with

$$\beta = 100, \quad R_E = 1 \text{ k}\Omega, \quad R_L = 1 \text{ k}\Omega, \quad R_B = 50 \text{ k}\Omega, \quad R_S = 2 \text{ k}\Omega.$$

The load is AC-coupled, all relevant capacitors act as shorts at the measurement frequency, and r_o is neglected.

- (a) Calculate g_m , r_π , and the effective emitter AC load $R_E \parallel R_L$.
- (b) Find the base-to-emitter gain and input resistance seen by the source.
- (c) Determine the source-to-output gain.
- (d) Remove R_L and determine the output resistance using the effective base grounding resistance $R_S \parallel R_B$.
- (e) Explain why the follower is useful despite having voltage gain below unity.

Q26. Common-base and cascode analysis — Advanced numerical

A CB amplifier has its base at AC ground and operates at

$$I_{CQ} = 1 \text{ mA}, \quad \beta = 100, \quad R_C = 4.7 \text{ k}\Omega, \quad R_L = 10 \text{ k}\Omega.$$

Neglect r_o .

- (a) Calculate its emitter input resistance and emitter-to-collector voltage gain.
- (b) A voltage source with $R_S = 50 \Omega$ drives the emitter. Find the source-to-output gain.
- (c) Now consider an idealized CE-CB cascode in which both transistors carry 1 mA , the lower emitter is AC-grounded, and the same collector load is used. Assume the upper transistor transfers the lower transistor's signal collector current with unity gain. Estimate the lower-base-to-output voltage gain.

- (d) State the sign of the cascode gain and explain how its input differs from that of the CB stage alone.

Q27. Two-stage cascade and loading — Advanced numerical

Two CE stages are capacitively coupled. At midband, their emitters are AC-grounded and r_o is neglected.

Stage 1 has

$$g_{m1} = 40 \text{ mS}, \quad R_{C1} = 4.7 \text{ k}\Omega, \quad R_{\text{in},1} = 10 \text{ k}\Omega.$$

Stage 2 has

$$g_{m2} = 20 \text{ mS}, \quad R_{C2} = 3.3 \text{ k}\Omega, \quad R_{\text{in},2} = 5 \text{ k}\Omega.$$

A source with $R_S = 1 \text{ k}\Omega$ drives stage 1, and the final load is $R_L = 10 \text{ k}\Omega$.

- Calculate the loaded gain of each stage.
- Calculate the overall source-to-final-output gain and its phase relationship.
- Compare this with the product of the two unloaded stage gains, ignoring source attenuation. Explain the difference.
- If the available undistorted final-output peak swing is 2 V, estimate the largest source peak amplitude allowed by the calculated overall gain. Assume the first stage remains within its own swing limits.

Q28. Low-frequency RC-coupled response — Frequency-analysis numerical

For an RC-coupled amplifier, use the following isolated-pole approximations:

$$R_S = 1 \text{ k}\Omega, \quad R_{\text{in}} = 10 \text{ k}\Omega, \quad C_{\text{in}} = 1 \mu\text{F},$$

$$R_{\text{out}} = 2 \text{ k}\Omega, \quad R_L = 10 \text{ k}\Omega, \quad C_{\text{out}} = 1 \mu\text{F}.$$

The emitter-bypass capacitor is $C_E = 100 \mu\text{F}$, and its effective resistance, with that capacitor removed and independent sources set to zero, is given as 100Ω .

- Estimate the input-coupling corner using $R_S + R_{\text{in}}$.
- Estimate the output-coupling corner using $R_{\text{out}} + R_L$.
- Estimate the emitter-bypass corner using the given resistance.
- Compare the three corner frequencies. Is a single clearly dominant low-frequency pole present?
- Explain why simply selecting the largest corner is only an approximation when several corners are close. Discuss the expected effect of increasing C_{in} by a factor of ten.

Q29. High-frequency response and the Miller effect — Advanced numerical

A CE amplifier has an AC-grounded emitter and

$$g_m = 40 \text{ mS}, \quad r_\pi = 2.5 \text{ k}\Omega, \quad R_C \parallel R_L = 2 \text{ k}\Omega,$$

$$R_S = 1 \text{ k}\Omega, \quad R_B = 50 \text{ k}\Omega, \quad C_\pi = 20 \text{ pF}, \quad C_\mu = 2 \text{ pF}.$$

An additional 10 pF capacitance connects the collector to AC ground. Neglect r_o .

Use the midband gain for an approximate Miller analysis.

- (a) Calculate the base-to-collector midband voltage gain.
- (b) Estimate the total input capacitance:

$$C_{\text{in,HF}} \approx C_{\pi} + C_{\mu}(1 - A_v).$$

- (c) Calculate the approximate input-pole frequency using $R_S \parallel R_B \parallel r_{\pi}$.
- (d) Estimate the output capacitance using

$$C_{\text{out,HF}} \approx 10 \text{ pF} + C_{\mu}(1 - 1/A_v),$$

and find the approximate output-pole frequency.

- (e) Identify the likely dominant upper-frequency pole. Explain why replacing the CE stage by a cascode can reduce the Miller multiplication at the input.

Q30. Class A and Class B power calculations — Power-amplifier numerical

- (a) An ideal series-fed Class A stage has $V_{CC} = 12 \text{ V}$, $R_C = 100 \Omega$, and an ideal midpoint Q-point. Neglect saturation voltage. Find I_{CQ} , DC supply power, maximum sinusoidal AC power in R_C , and maximum efficiency.
- (b) An ideal transformer-coupled Class A stage has $V_{CC} = 12 \text{ V}$ and $I_{CQ} = 60 \text{ mA}$. Find the reflected collector load needed to permit simultaneous peak swings $V_m = V_{CC}$ and $I_m = I_{CQ}$. Calculate maximum output power and efficiency.
- (c) An ideal complementary Class B output stage uses dual supplies of $+12 \text{ V}$ and -12 V and drives an 8Ω load with $v_o(t) = 10 \sin \omega t \text{ V}$. Find peak load current, average current drawn from each supply, total DC input power, AC output power, and efficiency.
- (d) Explain why the efficiencies in these cases differ. State which ideal stage draws zero supply current at idle.

Laboratory preparation checklist

For each amplifier studied, be prepared to:

- identify the transistor terminals and check their pinout;
- measure V_B , V_E , and V_C before applying a signal;
- calculate V_{BE} and V_{CE} from measured voltages;
- verify the operating region and compare the measured Q-point with the predicted value;
- distinguish DC offset from AC signal amplitude on an oscilloscope;
- measure gain and phase using the actual amplifier input voltage;
- identify clipping and reduce the input before measuring gain;
- measure a frequency-response curve at a sufficiently small input amplitude;
- explain differences caused by loading, component tolerances, transistor variation, and capacitor approximations.